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MAM1019H

Hand-in 7

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Answers

1. (a) False
(b) False
(c) True
(d) False
2. $\leq$ on $\mathbb{N}_0$ is defined by: $m \leq n \iff \exists z \in \mathbb{N}_0$, such that $m + z = n$. So $2 \leq 4 \iff \exists z \in \mathbb{N}_0$, such that $2 + z = 4$. Now $2 + 2 = 4$ and $2 \in \mathbb{N}_0$ and so $2 \leq 4$ (taking $z = 2$)
 $<$ on $\mathbb{N}_0$ is defined by: $m < n \iff m \leq n$ and $m \neq n$. So $4 < 10 \iff 4 \leq 10$ and $4 \neq 10 \iff \exists z \in \mathbb{N}_0, 4 + z = 10$ and $4 \neq 10$. Now $4 + 6 = 10$ and $6 \in \mathbb{N}_0$ so $4 \leq 10$. Moreover, $4 \neq 10$ so $4 < 10$.
3. $m \geq n \iff n \leq m$. Since $n \leq m \iff \exists z \in \mathbb{N}_0, n + z = m$, that means $m \geq n \iff \exists z \in \mathbb{N}_0, n + z = m$.
We use this to define $>$ on $\mathbb{N}_0$. So $m > n \iff m \geq n$ and $m \neq n$
4. Suppose $m \leq n$. That is, $\exists z \in \mathbb{N}_0$ such that $m + z = n$. Thus, $s(m + z) = s(n) \Rightarrow m + s(z) = s(n)$ by (A2) with $n = m$ and $m = z$ (in the axiom definition). Now $s(z) \in \mathbb{N}_0$, since $z \in \mathbb{N}_0$. Thus $m \leq s(n)$